# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

### *Introduction; Electrode potential definition; Electrochemical Cells*

**1.** Strong electrolytes are those which:

(A) dissolve readily in water

(B) conduct electricity

(C) dissociate into ions even at high concentration

(D) all of the above

**CEC087**

### *Galvanic Cell, Measurements of Electrode Potential, Representation of cell*

**2.** Correct cell reaction for the cell representation Pt | $Fe^{2\oplus}$, $Fe^{3\oplus}$ | | $MnO_4^{\ominus}$, $Mn^{2\oplus}$, $H^{\oplus}$ | Pt is :

(A) $MnO_4^{\ominus}(aq) + Fe^{2\oplus}(aq) + H^{\oplus}(aq) \longrightarrow Mn^{2\oplus}(aq) + Fe^{3\oplus}(aq) + H_2O(l)$

(B) $MnO_4^{\ominus}(aq) + 5Fe^{2\oplus}(aq) + 8H^{\oplus}(aq) \longrightarrow Mn^{2\oplus}(aq) + 5Fe^{3\oplus}(aq) + 4H_2O(l)$

(C) $MnO_4^{\ominus}(aq) + Fe^{2\oplus}(aq) \longrightarrow Mn^{2\oplus}(aq) + Fe^{3\oplus}(aq)$

(D) None of the above

**CEC110**

**3.** In galvanic cell, the salt bridge is used to

(A) Complete the circuit

(B) Reduce the electric resistance in the cell

(C) Separate cathode from anode

(D) Carry salts for the chemical reaction

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**CEC098**

### *Redox electrode*

**4.** In which of the following reaction $H_2O_2$ acts as reducing agent :

(A) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$

(B) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$

(C) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$

(D) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

**CEC091**

### *Cell Potential and Its relation with $\Delta G$*

**5.** Given $E^{o}_{Cl_2/Cl^{\ominus}} = 1.36\,V, E^{o}_{Cr^{3\oplus}/Cr} = -0.74\,V$ ; $E^{o}_{Cr_2O_7^{2\ominus}/Cr^{3\oplus}} = 1.33\,V, E^{o}_{MnO_4^{\ominus}/Mn^{2\oplus}} = 1.51\,V$ .

Among the following, the strongest reducing agent is :

(A) Cr

(B) $Mn^{2\oplus}$

(C) $Cr^{3\oplus}$

(D) $Cl^{\ominus}$

**CEC101**

**6.** In $Cu-Zn$ cell

(A) Reduction occurs at the copper cathode

(B) Oxidation occurs at the copper cathode

(C) Reduction occurs at the anode

(D) Chemical energy is converted to light energy

**CEC095**



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**7.** Consider the following reduction processes :

$Zn^{2\oplus}(aq) + 2e^{\ominus} \rightarrow Zn(s)$; $E° = -0.76$ V  $Ca^{2\oplus}(aq) + 2e^{\ominus} \rightarrow Ca(s)$; $E° = -2.87$ V

$Mg^{2\oplus}(aq) + 2e^{\ominus} \rightarrow Mg(s)$; $E° = -2.36$ V  $Ni^{2\oplus}(aq) + 2e^{\ominus} \rightarrow Ni(s)$; $E° = -0.25$ V

The reducing power of the metals increases in the order :

(A) Ca < Zn < Mg < Ni  (B) Ni < Zn < Mg < Ca  (C) Zn < Mg < Ni < Ca  (D) Ca < Mg < Zn < Ni

**CEC115**

## *Nernst equation and application of Nernst equation*

**8.** The emf of the cell,

$Tl_{(s)}|Tl^{\oplus}_{(aq)}$ (0.0001 M)|| $Cu^{2\oplus}_{(aq)}$(0.01M) | $Cu_{(s)}$ is 0.83 V

The emf of this cell will be increased by :–

(A) Increasing the concentration of $Cu^{2\oplus}$ ions  (B) Decreasing the concentration of $Tl^{\oplus}$

(C) Increasing the concentration of both  (D) (A) & (B) both

**CEC119**

**9.** Given : $E^0_{Cr^{3\oplus}/Cr} = -0.72$ V, $E^0_{Fe^{2\oplus}/Fe} = -0.42$V. The potential for the cell

$Cr_{(s)}$ | $Cr^{3\oplus}_{(aq)}$ (0.1 M) || $Fe^{2\oplus}_{(aq)}$ (0.01 M)|$Fe_{(s)}$ is

(A) 0.26 V  (B) 0.339 V  (C) – 0.339 V  (D) – 0.26 V

**CEC123**

**10.** The standard emf of a galvanic cell involving cell reaction with n = 4 is found to be 0.295 V at 25°C. The equilibrium constant of the reaction would be,

(A) $1.0 \times 10^{20}$  (B) $2.0 \times 10^{11}$  (C) $4.0 \times 10^{12}$  (D) $1.0 \times 10^{2}$

**CEC125**

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**11.** The reduction potential of hydrogen half-cell will be negative if :

(A) $p(H_2) = 2$ atm $[H^{\oplus}] = 1.0$ M  (B) $p(H_2) = 2$ atm and $[H^{\oplus}] = 2.0$ M

(C) $p(H_2) = 1$ atm and $[H^{\oplus}] = 2.0$ M  (D) $p(H_2) = 1$ atm and $[H^{\oplus}] = 1.0$ M

**CEC127**

**12.** The cell Zn | $Zn^{2\oplus}_{(aq)}$(1M) || $Cu^{2\oplus}_{(aq)(1M)}$ | Cu ($E°_{cell} = 1.10$ V) was allowed to be completely discharged at 298 K. The relative concentration of $Zn^{2\oplus}$ to $Cu^{2\oplus}$, is :

(A) $9.65 \times 10^4$  (B) Antilog (24.08)  (C) 37.3  (D) $10^{37.3}$

**CEC144**

## *Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction*

**13.** The standard reduction potentials of $Cu^{2\oplus}|Cu$ and $Cu^{2\oplus}|Cu^{\oplus}$ are 0.337 and 0.153 V respectively. The standard electrode potential of $Cu^{\oplus}|Cu$ half cell is :

(A) 0.184 V  (B) 0.827 V  (C) 0.521 V  (D) 0.490 V

**CEC108**

**14.** For the electrochemical cell, M | $M^{\oplus}$ || $X^{\ominus}$ | X, $E^o_{M^{\oplus}|M} = 0.44$ V and $E^o_{X|X^{\ominus}} = 0.33$ V. From this data, one can deduce that

(A) $M + X \rightarrow M^{\oplus} + X^{\ominus}$ is the spontaneous reaction

(B) $M^{\oplus} + X^{\ominus} \rightarrow M + X$ is the spontaneous reaction

(C) $E_{cell} = 0.77$ V

(D) $E_{cell} = -0.77$ V

**CEC112**



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**15.** The standard reduction electrode potentials of four metal cations are -

$E^0_{A^{\oplus}/A} = -0.250$ V , $E^0_{B^{\oplus}/B} = -0.140$ V,

$E^0_{C^{\oplus}/C} = -0.126$ V , $E^0_{D^{\oplus}/D} = -0.402$ V

The metal that displaces $A^{\oplus}$ from its aqueous solution is :–

(A) B (B) C (C) D (D) None of the above

**CEC116**

**16.** $E°(Ni^{2\oplus}|Ni) = -0.25$ volt, $E°\ (Au^{3\oplus} | Au) = 1.50$ volt. The standard emf of the voltaic cell.

$Ni_{(s)} | Ni^{2\oplus}_{(aq)}\ (1.0\ M) || Au^{3\oplus}_{(aq)}\ (1.0\ M) | Au_{(s)}$ is :

(A) 1.25 volt (B) –1.75 volt

(C) 1.75 volt (D) 4.0 volt

**CEC142**

***Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes***

**17.** Resistance of 0.2 M solution of an electrolyte is 50 Ω. The specific conductance of the solution is 1.4 S $m^{-1}$. The resistance of 0.5 M solution of the same electrolyte is 280 Ω. The molar conductivity of 0.5 M solution of the electrolyte in S $m^2$ $mol^{-1}$ is :

(A) $5 \times 10^3$ (B) $5 \times 10^2$

(C) $5 \times 10^{-4}$ (D) $5 \times 10^{-3}$

**CEC165**

**18.** Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100 Ω. The conductivity of this solution is 1.29 $Sm^{-1}$. Resistance of the same cell when filled with 0.02M of the same solution is 520 Ω. The molar conductivity of 0.02M solution of the electrolyte will be.

(A) $124 \times 10^{-4}$ $Sm^2$ $mol^{-1}$ (B) $1240 \times 10^{-4}$ $Sm^2$ $mol^{-1}$

(C) $1.24 \times 10^4$ $Sm^2$ $mol^{-1}$ (D) $12.4 \times 10^{-4}$ $Sm^2$ $mol^{-1}$

**CEC197**

**19.** Electrolytic conduction differs from metallic conduction from the fact that in the former

(A) The resistance increases with increasing temperature

(B) The resistance decreases with increasing temperature

(C) The resistance remains constant with increasing temperature

(D) The resistance is independent of the length of the conductor

**CEC205**

***Equivalent Conductance; Molar conductance at infinite dilution or limiting molar conductivity***

**20.** The specific conductance of a 0.01 M solution of KCl is 0.0014 $ohm^{-1}$ $cm^{-1}$at 25° C. Its equivalent conductance ($cm^2$ $ohm^{-1}$ $equivalent^{-1}$) is :–

(A) 140 (B) 14 (C) 1.4 (D) 0.14

**CEC171**

**21.** Which of the following solutions of KCl has the lowest value of equivalent conductance ?

(A) 1 M (B) 0.1 M (C) 0.01 M (D) 0.001 M

**CEC207**



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**Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water**

**22.** At infinite dilution, the equivalent conductance of $CH_3COONa$, HCl and $CH_3COOH$ are 91, 426 and 391 mho $cm^2$ $eq^{-1}$ respectively at 25 °C. The equivalent conductance of NaCl at infinite dilution will be :

(A) 126 (B) 209 (C) 391 (D) 908

**CEC179**

**23.** Molar conductance of 0.1 M acetic acid is 7 $ohm^{-1}$ $cm^2$ $mol^{-1}$. If the molar cond. of acetic acid at infinite dilution is 380.8 $ohm^{-1}$ $cm^2$ $mol^{-1}$, the value of dissociation constant will be :

(A) $226 \times 10^{-5}$ mol $dm^{-3}$ (B) $1.66 \times 10^{-3}$ mol $dm^{-1}$

(C) $1.66 \times 10^{-2}$ mol $dm^{-3}$ (D) $3.442 \times 10^{-5}$ mol $dm^{-3}$

**CEC182**

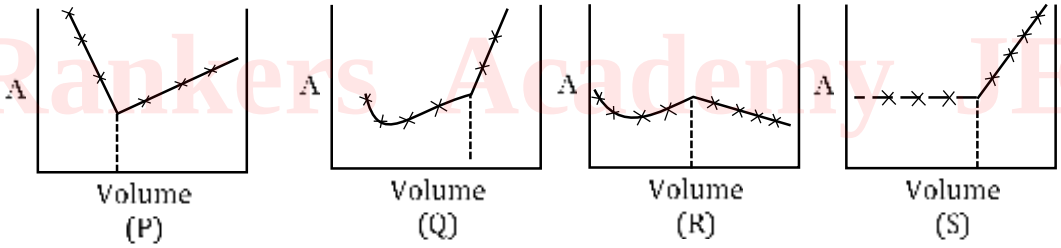
**24.** Equivalent conductance of 0.1 M HA(weak acid) solution is 10 $Scm^2 equivalent^{-1}$ and that at infinite dilution is 200 $Scm^2 equivalent^{-1}$ Hence pH of HA solution is

(A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7

**CEC201**

***Conductometric titration such as Acid base titrations and precipitation titration; Ionic mobility***

**25.** $AgNO_3$ was added to an aqueous KCl solution gradually and the conductivity of the solution was measured. The plot of conductance ($\Lambda$) versus the volume of $AgNO_3$ is -

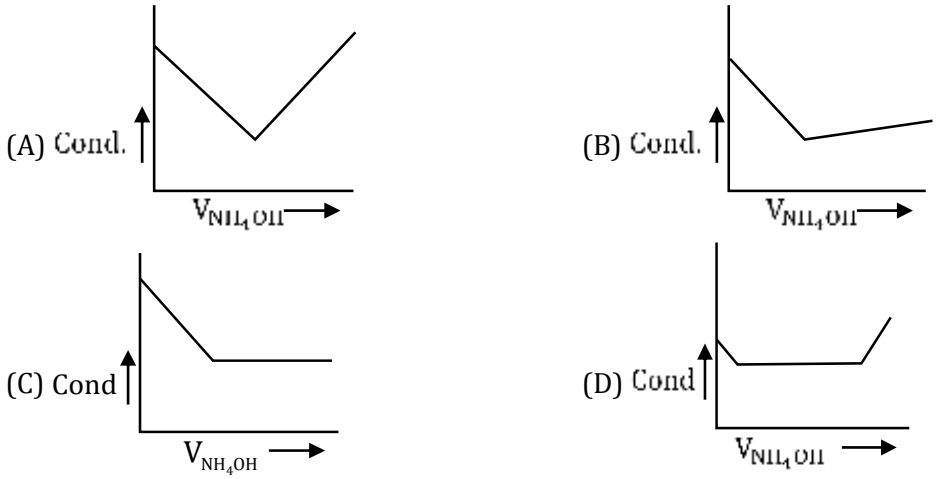



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(A) (P) (B) (Q) (C) (R) (D) (S)

**CEC185**

**26.** Which of the following graph truly represents the titration of HCl solution against $NH_4OH$ solution ?




**CEC188**



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## Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition

**27.** When an aqueous solution of lithium chloride is electrolysed using graphite electrodes
(A) $Cl_2$ is liberated at the anode.
(B) Li is deposited at the cathode
(C) as the current flows, pH of the solution remains constant
(D) as the current flows, pH of the solution decreases.

**CEC173**

## Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence

**28.** In the lead storage battery, the anode reaction is $Pb(s) + HSO_4^{\ominus} + H_2O \rightarrow PbSO_4(s) + H_3O^{\oplus} + 2e^{\ominus}$. How many grams of Pb will be used up to deliver 1 A for 100 h? (Pb = 208)
(A) 776 g (B) 388 g (C) 194 g (D) 0.1 g

**CEC140**

**29.** A solution of sodium sulphate in water is electrolysed using platinum electrodes. The products at cathode and anode are respectively:
(A) $H_2$, $O_2$ (B) $O_2$, $H_2$ (C) $O_2$, Na (D) $O_2$, $SO_2$

**CEC146**

**30.** The products formed when an aqueous solution of NaBr is electrolyzed in a cell having inert electrodes are :
(A) Na and $Br_2$ (B) Na and $O_2$
(C) $H_2$, $Br_2$ and NaOH (D) $H_2$ and $O_2$

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**CEC148**

**31.** How many coulomb of electricity are consumed when 100 mA current is passed through a solution of $AgNO_3$ for 30 minute during an electrolysis experiment :
(A) 108 (B) 18000 (C) 180 (D) 3000

**CEC150**

**32.** When an electric current is passed through acidulated water, 112 ml. of hydrogen gas at STP collects at the cathode in 965 second. The current passed, in ampere is :
(A) 1.0 (B) 0.5 (C) 0.1 (D) 2.0

**CEC152**

**33.** A current 0.1 A was passed for 965 second through a solution of $Cu^{\oplus}$ solution and 0.03175g of copper was deposited on the cathode. Calculate the current efficiency for the copper deposition. (molecular weight of Cu is 63.5)
(A) 79% (B) 39.5 %
(C) 63.25% (D) 50%

**CEC153**

**34.** The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[Ni(NO_3)_2]$ and chromium nitrate $[Cr(NO_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited is : (at. wt. of Ni = 59, at. wt. of Cr = 52)
(A) 0.1 g (B) 0.17 g (C) 0.3 g (D) 0.6 g

**CEC162**



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**35.** The ratio of weights of hydrogen and magnesium deposited by the same amount of electricity from aqueous $H_2SO_4$ and fused $MgSO_4$ are :

(A) 1 : 8 (B) 1 : 12 (C) 1 : 16 (D) None of these

**CEC175**

**36.** One mole of electron passes through each of the solution of $AgNO_3$, $CuSO_4$ and $AlCl_3$ when Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are

(A) 1 : 1 : 1 (B) 6 : 3 : 2 (C) 6 : 3 : 1 (D) 1 : 3 : 6

**CEC176**

**37.** 3.17 g., of a substance was deposited by the flow of 0.1 mole of electrons. The equivalent weight of the substance is :

(A) 3.17 (B) 0.317 (C) 317 (D) 31.7

**CEC213**

## 25. Primary batteries; Dry cell; Leclanche cell; Mercury cell; Button cell; secondary batteries; Lead storage battery; Nickel Cadmium cell

**38.** When a lead storage battery is discharged

(A) $SO_2$ is evolved (B) Lead sulphate is consumed

(C) Lead is formed (D) Sulphuric acid is consumed

**CEC133**

**39.** When lead storage battery is charged

(A) Lead dioxide dissolves

(B) Sulphuric acid is regenerated

(C) The lead electrode becomes coated with lead sulphate

(D) The amount of sulphuric acid decreases

**CEC136**

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# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

### Cell Potential and Its relation with ΔG

**1.** Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (A), (B), (C) and (D).

$Fe^{2\oplus}_{(aq)} + 2e^{\ominus} \rightleftharpoons Fe_{(s)}$ ; $E° = -0.44$ V

$Cu^{2\oplus}_{(aq)} + 2e^{\ominus} \rightleftharpoons Cu_{(s)}$ ; $E° = +0.34$ V

$Ag^{\oplus}_{(aq)} + e^{\ominus} \rightleftharpoons Ag_{(s)}$ ; $E° = +0.80$ V

I. Copper can displace iron from $FeSO_4$ solution.

II. Iron can displace copper from $CuSO_4$ solution.

III. Silver can displace copper from $CuSO_4$ solution.

IV. Iron can displace silver from $AgNO_3$ solution.

(A) I and II (B) II and III (C) II and IV (D) I and IV

**CEC199**

### Nernst equation and application of Nernst equation

**2.** An electrochemical cell is made by combining two half cells (X) $AgNO_3$(0.05) M | Ag(s) and (Y) $AgNO_3$ (0.01) M | Ag(s) Choose the correct statement

(A) The $E_{cell}$ can be increased if concentration of $AgNO_3$ in half cell X is increased

(B) The $E_{cell}$ will increase if concentration of $AgNO_3$ in cell Y is increased

(C) The $E_{cell}$ will increase if concentration of $AgNO_3$ in cell X is decreased

(D) The $E_{cell}$ will increase if concentration of $AgNO_3$ in cell Y is decreased

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**CEC236**

### Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

**3.** The specific conductance of saturated solution of silver bromide is $K(\Omega^{-1}\,cm^{-1})$. The limiting ionic conduction of $Ag^+$ and $Br^-$ ions are a & b respectively. The solubility of AgBr in g $lit^{-1}$ is :
(Molar mass of AgBr = 188g $mol^{-1}$)

(A) $K \times \frac{1000}{a-b}$ (B) $\frac{K}{a+b} \times 188$ (C) $\frac{K \times 1000 \times 188}{a+b}$ (D) $\frac{a+b}{K} \times \frac{1000}{188}$

**CEC237**

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question # 4**

The electrochemical cell shown below is a concentration cell.

$M \mid M^{2\oplus}$ (saturated solution of a sparingly soluble salt, $MX_2$) $\mid\mid M^{2\oplus}$ (0.001 mol $dm^{-3}$) | M

The emf of the cell depends on the difference in concentrations of $M^{2\oplus}$ ions at the two electrodes. The emf of the cell at 298 K is 0.059V.

**4.** The value of ΔG (kJ $mol^{-1}$) for the given cell is (take 1F = 96500 C $mol^{-1}$)

(A) –5.7 (B) 5.7

(C) 11.4 (D) –11.4

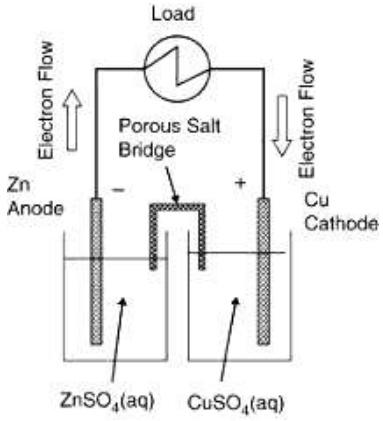
**CEC211**



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**Paragraph for Question 5 to 6**

The most popular electrochemical cell, Daniel cell was originally developed by the English chemist John F. Daniel. The general assembly of the cell is as given below




An undergraduate student setup a Daniel cell using 100 $cm^3$ of 0.100 M $CuSO_4$ and 100 $cm^3$ 0.100 M $ZnSO_4$ solution respectively. The two compartments are connected by suitable salt bridge.

(Given = $E^0_{Cu^{2+}/Cu}$ = 0.34 V, $E^0_{Zn^{2+}/Zn}$ = –0.76 V, $\frac{2.303RT}{F}$ = 0.06, log2 = 0.3)

**5.** A lab mate of the student asked her for some solid $CuCl_2$. While she was lining the bottle from a shelf, the lid of the bottle slipped and some amount of $CuCl_2$ fell in the $CuSO_4$ compartment at constant volume. She measured the emf of the cell again and found that it had increased by 9 mV. She used this data to calculate the amount of $CuCl_2$ that had split in the compartment. Calculate the mass of $CuCl_2$ in grams that had split into the Daniel cell? (Molecular mass of $CuCl_2$ = 135g)

(A) 1.35 (B) 13.5 (C) 27 (D) 2.7

**CEC238**

**6.** After making this calculation, the student discarded the solution from both the compartments and washed the cell thoroughly. She refilled the compartments with 100 $cm_3$ of the original $CuSO_4$ and $ZnSO_4$ solutions. A current of 0.965 A was drawn from the cell for a period of 20 min. What will be the cell potential after 20 min. ?

(A) 1.091 V (B) 1.109 V (C) 1.118 V (D) 1.082 V

**CEC239**

***Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction***

**Paragraph for Question 7 to 8**

Electrochemical techniques are often used to determine the solubilities of sparingly soluble salts. EMF of a cell is the linear function of logarithm of the concentration of ionic species. This enables us to determine even very small concentration of ions. An electrochemical cell is constructed with one half cell containng Zn(s) rod dipped in 0.2 M $Zn(NO_3)_2$ (aq.) solution and another half consisting Ag(s) rod immersed in 0.1M $AgNO_3$ solution. Each solution has a volume of 1L and temperature of 298 K $E^0_{Ag^+|Ag}$ = 0.80 V $E^0_{Zn^{2+}/Zn}$ = – 0.76 V

**7.** $K_{eq}$ for reaction given below is

$2n + 2Ag^+$ (aq.) $\rightleftharpoons$ $2Ag + Zn^{2+}$ (aq.)

(A) $10^{52}$ (B) $10^{-52}$ (C) $10^{26}$ (D) $10^{-26}$

**CEC240**



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**8.** EMF of the above cell reaction becomes 0.981 V and $[K^+] = 0.3$ M upon addition of KCl into $AgNO_3$ solution. $K_{sp}$(AgCl) & $Cl^-$ ion concentration in equilibrium with precipitated AgCl respectively are:

(A) $1 \times 10^{-20}$ $M^2$, 0.2 M (B) $2 \times 10^{-29}M^2$, 0.2 M

(C) $2 \times 10^{-11}$ $M^2$, 0.2 M (D) $2 \times 10^{-11}M^2$, $10^{-10}$ M

**CEC241**

## MATRIX MATCH TYPE QUESTION

**Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition**

**9.** Electrolysis of following aqueous solution proceed in electrolytic cell for small period of time.

| List-I (Aqueous solution) | | List-II (Products) | |
|---|---|---|---|
| (P) | $CH_3COOK$ (Pt electrodes) | (1) | Metal becomes deposited at cathode |
| (Q) | Dilute NaCl (Pt electrodes) | (2) | $H_2$ gas at cathode |
| (R) | $CuSO_4$ (Pt electrodes) | (3) | $H_2$ gas at anode |
| (S) | $Ni(NO_3)_2$ (Ni electrodes) | (4) | $O_2$ gas at anode |
| | | (5) | Concentration of electrolyte remain almost same |

| | P | Q | R | S |
|---|---|---|---|---|
| (1) | 1 | 4 | 2 | 4 |
| (2) | 2 | 4 | 2 | 4 |
| (3) | 2 | 2 | 4 | 5 |
| (4) | 2 | 1 | 4 | 5 |

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**CEC242**

**Primary batteries; Dry cell; Leclanche cell; Mercury cell; Button cell; secondary batteries; Lead storage battery; Nickel Cadmium cell**

**10.** Statement -1 : The voltage of mercury cell remains constant for long period of time.

Statement -2 : It is because net cell reaction does not involve active species.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

**CEC138**



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# EXERCISE (S)

## Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

### Nernst equation and application of Nernst equation

**1.** Pt| $\underset{1\text{ bar}}{H_2}$ |$H^{\oplus}$||$Ag^{\oplus}$ (0.1M)| Ag ; $E_{cell}$ = 0.86 volt

Find pH in anodic half cell

(Given $\frac{2.303RT}{F} = 0.06$ ; $E^{o}_{Ag^{\oplus}/Ag} = 0.8$ volt )

**CEC129**

**2.** $I_2(s)$ | $I^{\ominus}$ (0.1M) half-cell is connected to $H^{\oplus}_{(aq)}$ | $H_2$(1 atm) | Pt half-cell and its half-cell potential is found to be 0.7714 V. If $E^{0}_{I_2/I^{\ominus}} = 0.535V$ , the pH of $H^{\oplus}/H_2$ half-cell will be :-

[Give answer as a closest integer]

**CEC157**

### Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction

**3.** The equilibrium constant for the reaction $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ at 25°C is $2.18 \times 10^{y}$. The value of y is

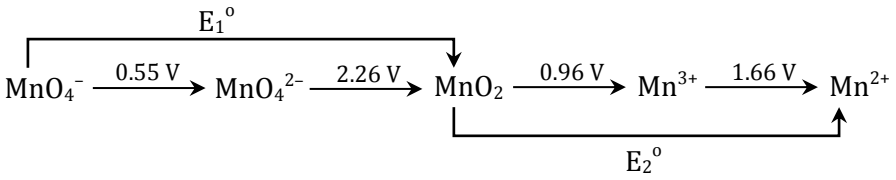
Given $E^0$ ($Fe|Fe^{2\oplus}$) = 0.44V, $E^0$ ($Cu|Cu^{2\oplus}$) = –0.337V. [$10^{26.3389} = 2.18 \times 10^{26}$]

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**CEC159**

### Latimer diagram

**4.** From the standard potential in acidic medium as shown in the following latimer diagram, the value of ($E_1° + E_2°$), in volts, is –




**CEC217**

### Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes

**5.** The molar conductivity of 0.1 M $CH_3COOH$ solution is 4 S $cm^2$ $mol^{-1}$. What is the resistivity of the solution ?

**CEC190**

### Equivalent Conductance; Molar conductance at infinite dilution or limiting molar conductivity

**6.** For 0.01N KCl, the resistivity 800 ohm cm. Find equivalent conductivity of KCl in S $cm^2$ $equivalent^{-1}$ ?

**CEC230**



For More Join: @IITJEE_Advanced

**Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water**

**7.** Conductivity of an aqueous solution of 0.1 M HX (a weak mono-protic acid) is $5 \times 10^{-4}$ $Sm^{-1}$.
Given : $\Lambda_m^{\infty}[H^{\oplus}] = 0.04\ Sm^2mol^{-1}$ ; $\Lambda_m^{\infty}[X^{\ominus}] = 0.01\ Sm^2mol^{-1}$
Find $pK_a$[HX].

**CEC202**

**8.** Saturated solution of AgCl at 25°C has specific conductance of $1.12\times10^{-6}$ $ohm^{-1}$ $cm^{-1}$. The $\lambda_{\infty}$ ($Ag^{\oplus}$) and $\lambda_{\infty}$($Cl^{\ominus}$) are 54 and 58 $ohm^{-1}$ $cm^2$/equivalent respectively. The solubility product of AgCl (at 25°C) is $10^{-y}$. The value of y is

**CEC203**

**9.** Molar conductance of $BaCl_2$, $H_2SO_4$ and HCl at infinite dilutions are 300 $\Omega^{-1}cm^2$/mole, 800 $\Omega^{-1}cm^2$/mole and 400 $\Omega^{-1}cm^2$/mole respectively. Equivalent conductance of $BaSO_4$ at infinite dilution is. (in S-$cm^2$/eq.)

**CEC232**

**10.** $\Lambda_m^o$ for $CH_3COOH$ based on the data [All data in S–$cm^2$ $mol^{-1}$].

| Electrolyte | $CaCl_2$ | $(CH_3COO)_2Ca$ | HCl |
|---|---|---|---|
| $\Lambda_m^o$ | 1400 | 800 | 700 |

**CEC231**

***Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence***

**11.** After electrolysis of NaCl solution with inert electrodes for a certain period of time. 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu was deposited in the copper voltameter in series with the NaCl electrolytic cell. Calculate the percentage yield of NaOH obtained.

**CEC155**

**12.** During electrolysis of NaCl, if 3 moles of NaCl are electrolysed then how much charge in faradays is required if current efficiency is 75% for the process?

**CEC227**

**13.** A current of 3.7A is passed for 6hrs. between Ni electrodes in 0.5L of 2M solution of $Ni(NO_3)_2$. What will be the molarity of $Ni(NO_3)_2$ solution at the end of electrolysis?

**CEC228**

## Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

***Nernst equation and application of Nernst equation***

**14.** At 25°C the value of $K_{eq}$ for the equilibrium $Fe^{3\oplus} + Ag \longrightarrow Fe^{2\oplus} + Ag^{\oplus}$ is 0.531 mol/litre. What is the value of $E^0_{Fe^{3\oplus}|Fe^{2\oplus}}$ ?

($E^0_{Ag^{\oplus}|Ag} = 0.799$ V ) [log (0.531 = – 0.27)] **CEC160**



For More Join: @IITJEE_Advanced

**15.** If for the half cell reactions

$Cu^{2+} + e^- \longrightarrow Cu^+$ $\quad E° = 0.15$ V

$Cu^{2+} + 2e^- \longrightarrow Cu$ $\quad E° = 0.34$ V

Calculate E° in volts of the half cell reaction

$Cu^+ + e^- \longrightarrow Cu$

and calculate $E^o$ in volts for disproportion of $Cu^+$

**CEC216**

**16.** Calculate potential in volts of reaction given below :

$2Ag^+(1M) + H_2(1bar) \rightarrow 2Ag(s) + 2H^+(1M)$

$E^o_{Ag^+|Ag} = 0.80$ V, $E^o_{H^+|H_2(Pt)} = 0.0$ V

**CEC219**

**17.** Equinormal Solutions of two weak acids, HA ($pK_a = 3$) and HB ($pK_a = 5$) are each placed in contact with equal pressure of hydrogen electrode at 25°C. When a cell is constructed by interconnecting them through a salt bridge, find the emf of the cell. (in volts)

**CEC221**

**18.** In two vessels each containing 500ml water, 0.5m mol of aniline ($K_b = 10^{-9}$) and 50mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf in volts of cell made by connecting them appropriately.

**CEC222**

**Thermodynamics of a cell; Temperature coefficient**

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**19.** For the cell reaction : $Hg_2Cl_2(s) + 2Ag(s) \rightarrow 2Hg(l) + 2AgCl(s)$ temperature coefficient of cell emf is found to be 0.02 $VK^{-1}$. Find $\Delta_r S°$ for cell reaction in kJ $mole^{-1}$

**CEC224**

**Calculation of Electrode potential of half cell and complete cell reaction; Equilibrium constant of the half cell and the full cell reaction**

**20.** If value of equilibrium constant for given reaction is $A \times 10^{12}$ then value of A will be :

$Fe^{2+} + Ce^{4+} \rightleftharpoons Fe^{3+} + Ce^{3+}$, [Given : $E^0_{Ce^{4+}|Ce^{3+}} = 1.44$ V; $E^0_{Fe^{3+}|Fe^{2+}} = 0.68$ V ] [$10^{12.88} = 7.6 \times 10^{12}$]

**CEC223**

**Latimer diagram**

**21.** Find $E^o$ at 298 K for the reaction in volts,

$Cu^{2+}(aq) + In^{2+}(aq) \rightleftharpoons Cu^+(aq) + In^{3+}(aq)$

Given that $E^o_{Cu^{2+}|Cu^+} = 0.15V$ , $E^o_{In^{3+}|In^+} = -0.42V$ , $E^o_{In^{2+}|In^+} = -0.40V$

**CEC218**

**Conductance of Electrolytic Solutions; Resistance; Molar conductance; Lambda Conductivity or specific conductance; Kappa; Cell constant; Type of electrolytes; Strong and Weak Electrolytes**

**22.** A solution containing 2.08 g of anhydrous barium chloride in 500 $cm^3$ of water has a specific conductivity 0.005 $ohm^{-1}cm^{-1}$. What are molar and equivalent conductivities of this solution in mho $cm^2$ $mol^{-1}$ and mho $cm^2$ $equivalent^{-1}$ respectively.

**CEC229**



For More Join: @IITJEE_Advanced

**Statement and application of Kohlraush law for calculating the degree of dissociation and molar conductance at infinite dilution for weak electrolytes; Solubility of sparingly soluble salt precipitates; dissociation or ionisation constant of water**

**23.** Calculate standard electrode reduction potential in volts when silver wire coated with a layer of silver iodide is placed in solution of KI at 25°C.

Given : $E°_{Ag^+/Ag} = 0.8$ V ; $K_{sp}(AgI) = 10^{-16}$ ; $\frac{2.303\,RT}{F} = 0.06$

**CEC220**

**24.** Specific conductance of a saturated solution of AgBr is $8.075 \times 10^{-7}$ $ohm^{-1}cm^{-1}$ at $25^0$C. Specific conductance of pure water at 25°C is $0.075 \times 10^{-7}$ $ohm^{-1}$ $cm^{-2}$. $\Lambda_m^\infty$ for KBr, $AgNO_3$ and $KNO_3$ are 140, 130, 110 (S $cm^2$ $mol^{-1}$) respectively. If the solubility of AgBr is ($A \times 10^{-4}$ gm/litre) then value of A will be ? [Atomic mass : Ag = 108 , Br = 80]

**CEC233**

**Electrolytic cell; Electrolysis using inert electrodes; Active electrodes; electrolysis of aqueous NaCl and a few more examples of electrolysis; Sequence of deposition**

**25.** Calculate mole of electrons lost or gained during electrolysis of

(a) 3.55 gm of $Cl^-$ ions (b) 1 gm $Cu^{2+}$ ions

(c) 2.7 gm of $Al^{3+}$ ions

[Atomic mass : Cu = 63.5 , Al = 27, Cl = 35.5]

**CEC225**

**Faraday's First and Second Law of Electrolysis; Electrochemical Equivalent(Z), Law of chemical equivalence**

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**26.** How many faradays of electricity are involved in each of the case

(a) 0.25 mole $Al^{3+}$ is converted to Al.

(b) 27.6 gm of $SO_3$ is converted to $SO_3^{2-}$

(c) The $Cu^{2+}$ in 1100 ml of 0.5 M $Cu^{2+}$ is converted to Cu.

**CEC226**



For More Join: @IITJEE_Advanced

# EXERCISE – JEE (Main) PYQs

**1.** The standard electrode potential of $M^+/M$ in aqueous solution does not depend on

**[JEE (Main) 2023]**

(1) Ionisation of a solid metal atom
(2) Sublimation of a solid metal
(3) Ionisation of a gaseous metal atom
(4) Hydration of a gaseous metal ion

**CEC243**

**2.** The reaction **[JEE (Main) 2023]**

$$\frac{1}{2}H_2(g) + AgCl(s) \rightleftharpoons H^+(aq) + Cl^-(aq) + Ag(s)$$

occurs in which of the given galvanic cell.

(1) $Pt|H_2(g)|KCl(sol^n)|AgCl(s)|Ag$
(2) $Pt|H_2(g)|HCl(sol^n)|AgCl(s)|Ag$
(3) $Ag|AgCl(s)|KCl(sol^n)|AgCl(s)|Ag$
(4) $Pt|H_2(g)|HCl(sol^n)|AgNO_3(sol^n)|Ag$

**CEC244**

**3.** Standard electrode potentials for a few half cells are mentioned below: **[JEE (Main) 2025]**

$E^o_{Cu^{2+}/Cu} = 0.34V, E^o_{Zn^{2+}/Zn} = -0.76V$

$E^o_{Ag^+/Ag} = 0.80V, E^o_{Mg^{2+}/Mg} = -2.37V$

Which one of the following cells gives the most negative value of $\Delta G°$ ?

(1) $Zn|Zn^{2+}(1M)||Ag^+(1M)|Ag$
(2) $Zn|Zn^{2+}(1M)||Mg^{2+}(1M)|Mg$
(3) $Ag|Ag^+(1M)||Mg^{2+}(1M)|Mg$
(4) $Cu|Cu^{2+}(1M)||Ag^+(1M)|Ag$

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**CEC293**

**4.** Based on the data given below: **[JEE (Main) 2025]**

$E^0_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33V \quad E^0_{Cl_2/Cl^{(-)}} = 1.36V$

$E^0_{MnO_4^-/Mn^{2+}} = 1.51V \quad E^0_{Cr^{3+}/Cr} = -0.74V$

the strongest reducing agent is :

(1) $Mn^{2+}$
(2) Cr
(3) $MnO_4^-$
(4) $Cl^-$

**CEC294**

**5.** The standard reduction potentials at 298 K for the following half cells are given below :

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, $E° = 1.33V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $E° = -0.04V$

$Ni^{2+}(aq) + 2e^- \rightarrow Ni$ $E° = -0.25V$

$Ag^+(aq) + e^- \rightarrow Ag$ $E° = 0.80V$

$Au^{3+}(aq) + 3e^- \rightarrow Au$ $E° = 1.40V$

Consider the given electrochemical reactions,

The number of metal(s) which will be oxidized be $Cr_2O_7^{2-}$, in aqueous solution is _____.

**[JEE (Main) 2024]**

**CEC246**



For More Join: @IITJEE_Advanced

**6.** The standard reduction potential at 298 K for the following half cells are given below :-

$NO_3^- + 4H^+ + 3e^- \rightarrow NO(g) + 2H_2O$ $E^0 = 0.97V$

$V^{2+}(aq) + 2e^- \rightarrow V$ $E^0 = -1.19V$

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$ $E^0 = -0.04V$

$Ag^+(aq) + e^- \rightarrow Ag(s)$ $E^0 = 0.80V$

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$ $E^0 = 1.40V$ **[JEE (Main) 2023]**

The number of metal(s) which will be oxidized by $NO_3^-$ in aqueous solution is _____.

**CEC247**

**7.** For a Mg | $Mg^{2+}$ (aq) || $Ag^+$(aq) | Ag the correct Nernst Equation is : **[JEE (Main) 2025]**

(1) $E_{cell} = E^o_{cell} - \frac{RT}{2F}\ln\frac{[Ag^+]}{[Mg^{2+}]}$ (2) $E_{cell} = E^o_{cell} + \frac{RT}{2F}\ln\frac{[Ag^+]^2}{[Mg^{2+}]}$

(3) $E_{cell} = E^o_{cell} - \frac{RT}{2F}\ln\frac{[Mg^{2+}]}{[Ag^+]}$ (4) $E_{cell} = E^o_{cell} - \frac{RT}{2F}\ln\frac{[Ag^+]^2}{[Mg^{2+}]}$

**CEC295**

**8.** The emf of cell $Tl\left|\underset{(0.001M)}{Tl^+}\right|\left|\underset{(0.01M)}{Cu^{2+}}\right|Cu$ is 0.83 V at 298 K. It could be increased by :

(1) increasing concentration of $Tl^+$ ions

(2) increasing concentration of both $Tl^+$ and $Cu^{2+}$ ions

(3) decreasing concentration of both $Tl^+$ and $Cu^{2+}$ ions

(4) increasing concentration of $Cu^{2+}$ ions **[JEE (Main) 2024]**

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**CEC248**

**9.** The hydrogen electrode is dipped in a solution of pH = 3 at 25°C. The potential of the electrode will be –_____ × $10^{-2}$ V.

$\left(\frac{2.303RT}{F} = 0.059\,V\right)$ **[JEE (Main) 2024]**

**CEC249**

**10.** At what pH, given half cell $MnO_4^-$ (0.1 M) | $Mn^{2+}$ (0.001 M) will have electrode potential of 1.282 V ?

_____ (Nearest Integer)

Given $E^o_{MnO_4^-/Mn^{2+}}$ = 1.54 V, $\frac{2.303RT}{F} = 0.059V$ **[JEE (Main) 2023]**

**CEC250**

**11.** The electrode potential of the following half cell at 298 K.

X | $X^{2+}$ (0.001 M) || $Y^{2+}$ (0.01 M) | Y

is ____ × $10^{-2}$ V (Nearest integer).

Given : $E^0_{X^{2+}|X} = -2.36V$

$E^0_{Y^{2+}|Y} = +0.36V$ $\frac{2.303RT}{F} = 0.06V$ **[JEE (Main) 2023]**

**CEC252**



For More Join: @IITJEE_Advanced

**12.** $Pt(s)H_2(g)(1bar)|H^+(aq)(1M)||M^{3+}(aq),M^+(aq)|Pt(s)$

The $E_{cell}$ for the given cell is 0.1115 V at 298 K when $\frac{[M^+(aq)]}{[M^{3+}(aq)]}=10^a$

The value of a is _____.

Given : $E^{\theta}_{M^{3+}/M^+}=0.2$ V

$\frac{2.303\,RT}{F}=0.059$ V

[JEE (Main) 2023]

**CEC253**

**13.** At 298 K, a 1 litre solution containing 10 mmol of $Cr_2O_7^{2\ominus}$ and 100 mmol of $Cr^{3\oplus}$ shows a pH of 3.0.

Given : $Cr_2O_7^{2\ominus} \rightarrow Cr^{3\oplus}$; $E^0 = 1.330$ V and

$\frac{2.303\,RT}{F} = 0.059$ V

The potential for the half cell reaction is $x \times 10^{-3}$ V. The value of x is _____. [JEE (Main) 2023]

**CEC049**

**14.** The standard electrode potential $E^0$ and its temperature coefficient $\left(\frac{dE^0}{dT}\right)$ for a cell are 2V and $-5 \times 10^{-4}$ $VK^{-1}$ at 300 K respectively. The cell reaction is

$Zn(s) + Cu^{2\oplus}(aq) \rightarrow Zn^{2\oplus}(aq) + Cu(s)$

The standard reaction enthalpy $\left(\Delta_rH^0\right)$ at 300 K in kJ $mol^{-1}$ is,

[Use : R = 8 $JK^{-1}$ $mol^{-1}$ and F = 96,000 $Cmol^{-1}$] [JEE (Main) 2019]

(1) –412.8 (2) –384.0 (3) 206.4 (4) 192.0

**CEC041**

**15.** For the given cell [JEE (Main) 2025]

$Fe^{2+}(eq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

The standard cell potential of the above reaction is Given :

$Ag^+ + e^- \rightarrow Ag$ $E^0 = xV$

$Fe^{2+} + 2e^- \rightarrow Fe$ $E^0 = y$ V

$Fe^{3+} + 3e^- \rightarrow Fe$ $E^0 = zV$

(1) x + y – z (2) x + 2y – 3z (3) y – 2x (4) x + 2y

**CEC296**

**16.** The potential for the given half cell at 298K is

(-)..... $\times 10^{-2}$ V.

$2H^+_{(aq)} + 2e^- \rightarrow H_2(g)$

$[H^+] = 1M$, $P_{H_2} = 2$ atm

(Given: 2.303 RT/F = 0.06 V, log2 = 0.3) [JEE (Main) 2024]

**CEC254**



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For More Join: @IITJEE_Advanced

**17.** For the electro chemical cell

$M|M^{2+}||X|X^{2-}$

If $E^0_{\left(M^{2+}/M\right)} = 0.46\,V$ and $E^0_{\left(X/X^{2-}\right)} = 0.34V$. Which of the following is **correct** ?

(1) $E_{cell} = -0.80$ V

(2) $M + X \rightarrow M^2 + X^{2-}$ is a spontaneous reaction

(3) $M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction

(4) $E_{cell} = 0.80$ V

[JEE (Main) 2024]

**CEC255**

**18.** Consider the following redox reaction :

$$MnO_4^- + H^+ + H_2C_2O_4 \rightleftharpoons Mn^{2+} + H_2O + CO_2$$

The standard reduction potentials are given as below $\left(E^\circ_{red}\right)$

$E^\circ_{MnO_4^-/Mn^{2+}} = +1.51V$

$E^\circ_{CO_2/H_2C_2O_4} = -0.49V$

If the equilibrium constant of the above reaction is given as $K_{eq} = 10^x$, then the value of x = _____ (nearest integer)

[JEE (Main) 2024]

**CEC256**

**19.** The equilibrium constant for the reaction

$$Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$$

is $1 \times 10^{20}$ at 298 K. The magnitude of standard electrode potential of $Sn/Sn^{2+}$ if $E^o_{Zn^{2+}/Zn} = -0.76$ V is _____ $\times 10^{-2}$ V.

(Nearest integer) Given : $\frac{2.303RT}{F} = 0.059V$

[JEE (Main) 2023]

**CEC259**

**20.** The logarithm of equilibrium constant for the reaction $Pd^{2+} + 4Cl^- \rightleftharpoons PdCl_4^{2-}$ is_____.

(Nearest integer)

Given : $\frac{2.303RT}{F} = 0.06V$

$Pd^{2+}_{(aq)} + 2e^- \rightleftharpoons Pd(s) \quad E^o = 0.83V$

$PdCl_4^{2-}(aq) + 2e^- \rightleftharpoons Pd(s) + 4Cl^-(aq)$

$E^o = 0.65V$

[JEE (Main) 2023]

**CEC260**

**21.** $FeO_4^{2-} \xrightarrow{+2.0\,V} Fe^{3+} \xrightarrow{0.8V} Fe^{2+} \xrightarrow{-0.5V} Fe^0$ [JEE (Main) 2025]

In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E^\Theta_{FeO_4^{2-}/Fe^{2+}}$ is

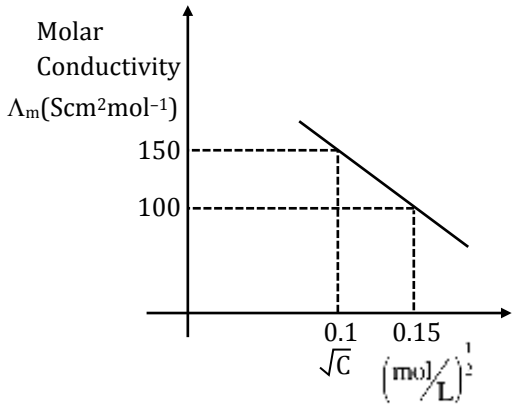
(1) 1.7 V (2) 1.2 V (3) 2.1 V (4) 1.4 V

**CEC297**



For More Join: @IITJEE_Advanced

**22.** Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.

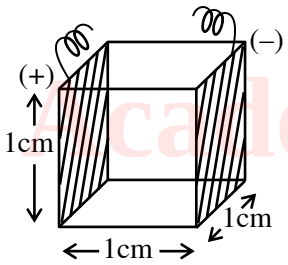



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is 100Ω, then the resistance of the same cell with the dilute solution is 'x'Ω.

The value of x is _____ (Nearest integer) **[JEE (Main) 2025]**

**CEC298**

**23.** A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will -




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(1) increase sharply
(2) remain same or can not be measured accurately
(3) decrease sharply
(4) depend upon type of electrolyte

**[JEE (Main) 2024]**

**CEC261**

**24.** The values of conductivity of some materials at 298.15 K in $Sm^{-1}$ are $2.1\times10^{3}$, $1.0\times10^{-16}$, $1.2\times10$, $3.91$, $1.5\times10^{-2}$, $1\times10^{-7}$, $1.0\times10^{3}$. The number of conductors among the materials is _____.

**[JEE (Main) 2024]**

**CEC262**

**25.** The resistivity of a 0.8 M solution of an electrolyte is $5\times10^{-3}$ Ωcm. Its molar conductivity is _____ $\times10^{4}\ \Omega^{-1}\ cm^{2}\ mol^{-1}$.

(Nearest integer) **[JEE (Main) 2023]**

**CEC263**



For More Join: @IITJEE_Advanced

**26.** The number of correct statements from the following is _____. **[JEE (Main) 2023]**

(A) Conductivity always decreases with decrease in concentration for both strong and weak electrolytes.

(B) The number of ions per unit volume that carry current in a solution increases on dilution.

(C) Molar conductivity increases with decrease in concentration.

(D) The variation in molar conductivity is different for strong and weak electrolytes.

(E) For weak electrolytes, the change in molar conductivity with dilution is due to decrease in degree of dissociation.

**CEC264**

**27.** The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed? **[JEE (Main) 2025]**

(1) A small decrease in molar conductivity is observed at infinite dilution.

(2) A small increase in molar conductivity is observed at infinite dilution.

(3) Molar conductivity increases sharply with increase in concentration.

(4) Molar conductivity decreases sharply with increase in concentration.

**CEC299**

**28.** For a strong electrolyte, a plot of molar conductivity against $(\text{concentration})^{1/2}$ is a straight line, with a negative slope, the correct unit for the slope is

(1) $S\ cm^2\ mol^{-3/2}\ L^{1/2}$ (2) $S\ cm^2\ mol^{-1}\ L^{1/2}$

(3) $S\ cm^2\ mol^{-3/2}\ L$ (4) $S\ cm^2\ mol^{-3/2}\ L^{-1/2}$

**[JEE (Main) 2024]**

**CEC265**

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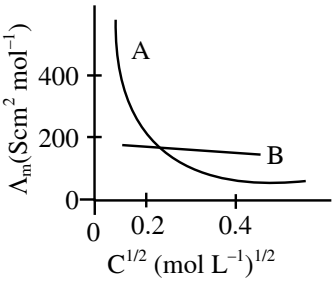
**29.** Molar ionic conductivities of divalent cation and anion are 57 $S\ cm^2\ mol^{-1}$ and 73 $S\ cm^2\ mol^{-1}$ respectively. The molar conductivity of solution of an electrolyte with the above cation and anion will be : **[JEE (Main) 2024]**

(1) 65 $S\ cm^2\ mol^{-1}$ (2) 130 $S\ cm^2\ mol^{-1}$

(3) 187 $S\ cm^2\ mol^{-1}$ (4) 260 $S\ cm^2\ mol^{-1}$

**CEC266**

**30.** The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are :




| | A | B |
|---|---|---|
| (1) | Weak electrolyte | weak electrolyte |
| (2) | Strong electrolyte | strong electrolyte |
| (3) | Weak electrolyte | strong electrolyte |
| (4) | Strong electrolyte | weak electrolyte |

**[JEE (Main) 2024]**

**CEC267**



For More Join: @IITJEE_Advanced

**31.** Which out of the following is a correct equation to show change in molar conductivity with respect to concentration for a weak electrolyte, if the symbols carry their usual meaning :

(1) $\Lambda^2{}_mC - K_a\Lambda_m^{\circ 2} + K_a\Lambda_m\Lambda^\circ{}_m = 0$

(2) $\Lambda_m - \Lambda^\circ{}_m + AC^{\frac{1}{2}} = 0$

(3) $\Lambda_m - \Lambda^\circ{}_m - AC^{\frac{1}{2}} = 0$

(4) $\Lambda^2{}_mC + K_a\Lambda_m^{\circ 2} - K_a\Lambda_m\Lambda^\circ{}_m = 0$

[JEE (Main) 2024]

**CEC268**

**32.** At 298 K, the standard reduction potential for $Cu^{2+}$ / Cu electrode is 0.34 V.

Given : $K_{sp}$ $Cu(OH)_2 = 1 \times 10^{-20}$

Take $\frac{2.303RT}{F} = 0.059V$

The reduction potential at pH = 14 for the above couple is $(-)x \times 10^{-2}$ V. The value of x is _____.

[JEE (Main) 2023]

**CEC269**

**33.** The specific conductance of 0.0025 M acetic acid is $5 \times 10^{-5}$ S $cm^{-1}$ at a certain temperature. The dissociation constant of acetic acid is _____ $\times 10^{-7}$. (Nearest integer)

Consider limiting molar conductivity of $CH_3COOH$ as 400 S $cm^2$ $mol^{-1}$

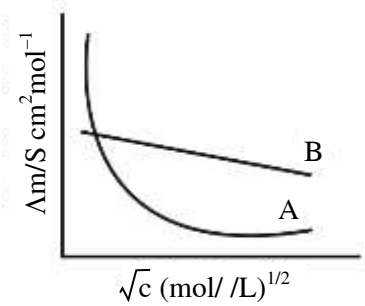
[JEE (Main) 2023]

**CEC270**

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**34.** Following figure shows dependence of molar conductance of two electrolytes on concentration.

$\Lambda^0_m$ is the limiting molar conductivity.




The number of **incorrect** statement(s) from the following is _____.

(A) $\Lambda^0_m$ for electrolyte A is obtained by extrapolation

(B) For electrolyte B, vx $\Lambda m$ vs $\sqrt{c}$ graph is a straight line with intercept equal to $\Lambda^0_m$

(C) At infinite dilution, the value of degree of dissociation approach zero for electrolyte B.

(D) $\Lambda^0_m$ for any electrolyte A or B can be calculated using $\lambda^o$ for individual ions.

[JEE (Main) 2023]

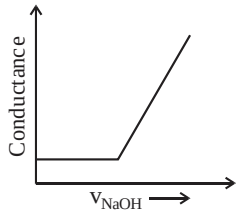
**CEC272**

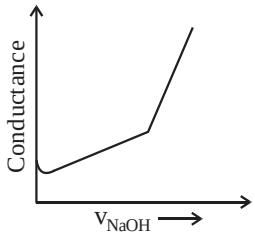


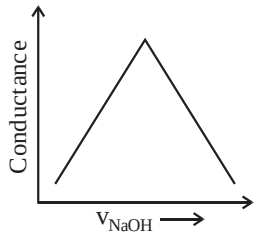
For More Join: @IITJEE_Advanced

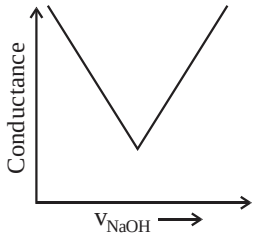
**35.** Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.

**[JEE (Main) 2023]**

(1) 


(2) 


(3) 


(4) 


**CEC050**

**36.** Which of the following electrolyte can be sued to obtain $H_2S_2O_8$ by the process of electrolysis?

**[JEE (Main) 2025]**

(1) Dilute solution of sodium sulphate

(2) Dilute solution of sulphuric acid

(3) Concentrated solution of sulphuric acid

(4) Acidified dilute solution of sodium sulphate.

**CEC300**

**37.** The product, which is not obtained during the electrolysis of brine solution is

**[JEE (Main) 2023]**

(1) NaOH (2) $Cl_2$

(3) $H_2$ (4) HCl

**CEC274**

**38.** Which one of the following statements is correct for electrolysis of brine solution?

**[JEE (Main) 2023]**

(1) $Cl_2$ is formed at cathode (2) $O_2$ is formed at cathode

(3) $H_2$ is formed at anode (4) $OH^-$ is formed at cathode

**CEC275**

**39.** A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2A. The amount of the aluminium deposited at the cathode is ___ .

[Given : molar mass of aluminium and chlorine are 27 g $mol^{-1}$ and 35.5 g $mol^{-1}$ respectively, Faraday constant = 96500 C $mol^{-1}$].

**[JEE (Main) 2025]**

(1) 1.660 g (2) 1.007 g

(3) 0.336 g (4) 0.441 g

**CEC301**

**40.** Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12. The current in Amperes used for the given electrolysis is ___. (Nearest integer).

**[JEE (Main) 2025]**

**CEC302**



For More Join: @IITJEE_Advanced

**41.** The mass of silver (Molar mass of Ag : 108 $gmol^{-1}$) displaced by a quantity of electricity which displaces 5600 mL of $O_2$ at S.T.P. will be ____ g. **[JEE (Main) 2024]**

**CEC276**

**42.** The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is___ $\times 10^{-4}$ g.
(Atomic mass of zinc = 65.4 amu) **[JEE (Main) 2024]**

**CEC277**

**43.** 0.05 cm thick coating of silver is deposited on a plate of 0.05 $m^2$ area. The number of silver atoms deposited on plate are ____ $\times 10^{23}$. (At mass Ag = 108, d = 7.9 g $cm^{-3}$)

**[JEE (Main) 2024]**

**CEC279**

**44.** One Faraday of electricity liberates x $\times 10^{-1}$ gram atom of copper from copper sulphate, x is_____.

**[JEE (Main) 2024]**

**CEC280**

**45.** A metal surface of 100 $cm^2$ area has to be coated with nickel layer of thickness 0.001mm. A current of 2A was passed through a solution of $Ni(NO_3)_2$ for 'x' seconds to coat the desired layer. The value of x is ____ (Nearest integer)
($\rho_{Ni}$ (density of Nickel) is 10 $gmL^{-1}$, Molar mass of Nickel is 60 $gmol^{-1}$ $F = 96500\ C\ mol^{-1}$)

**[JEE (Main) 2023]**

**CEC282**

**46.** The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of $PbSO_4$ electrolyzed (in g) during the process is : **(Molar mass of $PbSO_4$ = 303 g $mol^{-1}$)**

**[JEE (Main) 2019]**

(1) 22.8 (2) 15.2 (3) 7.6 (4) 11.4

**CEC044**

**47.** When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :- **[JEE (Main) 2018]**

(1) 10.9 g (2) 98.1 g
(3) 109.0 g (4) 9.81 g

**CEC038**

**48.** The metals that are employed in the battery industries are

A. Fe B. Mn C. Ni
D. Cr E. Cd

Choose the correct answer from the options given below:
(1) B, C and E only (2) A, B, C, D and E
(3) A, B, C and D only (4) B, D and E only

**[JEE (Main) 2024]**

**CEC283**

**49.** Identify the factor from the following that does not affect electrolytic conductance of a solution.
(1) The nature of the electrolyte added. (2) The nature of the electrode used.
(3) Concentration of the electrolyte. (4) The nature of solvent used.

**[JEE (Main) 2024]**

**CEC284**



For More Join: @IITJEE_Advanced

**50.** For lead storage battery pick the correct statements

**A.** During charging of battery, $PbSO_4$ on anode is converted into $PbO_2$

**B.** During charging of battery, $PbSO_4$ on cathode is converted into $PbO_2$

**C.** Lead storage battery, consists of grid of lead packed with $PbO_2$ as anode

**D.** Lead storage battery has $\sim$ 38% solution of sulphuric acid as an electrolyte

Choose the correct answer from the options given below :

**[JEE (Main) 2023]**

(1) B, D only (2) B, C, D only (3) A, B, D only (4) B, C only

**CEC285**

**51.** Given below are two statements : **[JEE (Main) 2025]**

**Statement (I) :** Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

**Statement (II) :** The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both Statement I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Statement I is true but Statement II is false

**CEC303**

**52.** Match List I with List II

| **LIST-I (Cell)** | | **LIST-II (Use/Property/Reaction)** | |
|---|---|---|---|
| A. | Leclanche cell | I. | Converts energy of combustion into electrical energy |
| B. | Ni-Cd cell | II. | Does not involve any ion in solution and is used in hearing aids |
| C. | Fuel cell | III. | Rechargeable |
| D. | Mercury cell | IV. | Reaction at anode $Zn \rightarrow Zn^{2+} + 2e^-$ |

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV (2) A-III, B-I, C-IV, D-II

(3) A-IV, B-III, C-I, D-II (4) A-II, B-III, C-IV, D-I

**[JEE (Main) 2024]**

**CEC286**



For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Advanced) PYQs

**1.** In a galvanic cell, the salt bridge - **[JEE (Advanced) 2014]**

(A) Does not participate chemically in the cell reaction

(B) Stops the diffusion of ions from one electrode to another

(C) Is necessary for the occurrence of the cell reaction

(D) Ensures mixing of the two electrolytic solutions

**CEC073**

**2.** Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is

$$H_2(g) + O_2(g) \rightarrow H_2O(\ell).$$

The work derived from the cell on the consumption of $1.0 \times 10^{-3}$ mol of $H_2(g)$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulted container. What is the change in the temperature (in K) of the ideal gas ?

The standard reduction potentials for the two half-cells are given below.

$$O_2(g) + 4\,H^{\oplus}(aq.) + 4e^{\ominus} \rightarrow 2H_2O(\ell),\ E^\circ = 1.23\ V,$$

$$2H^{\oplus}(aq.) + 2e^{\ominus} \rightarrow H_2(g),\ E^\circ = 0.00V.$$

Use F = 96500 C $mol^{-1}$, R = 8.314 J $mol^{-1}K^{-1}$ **[JEE (Advanced) 2020]**

**CEC081**

**3.** For the following cell :

$Zn(s) \mid ZnSO_4(aq.) \parallel CuSO_4(aq.) \mid Cu(s)$

when the concentration of $Zn^{2\oplus}$ is 10 times the concentration of $Cu^{2\oplus}$, the expression for $\Delta G$ (in J $mol^{-1}$) is

[F is Faraday constant, R is gas constant, T is temperature, E°(cell) = 1.1V] **[JEE (Advanced) 2017]**

(A) 2.303 RT + 1.1F (B) 2.303 RT – 2.2F (C) 1.1 F (D) –2.2 F

**CEC078**

**4.** For the electrochemical cell,

$$Mg(s) \mid Mg^{2\oplus}(aq, 1\ M) \mid\mid Cu^{2\oplus}(aq, 1M) \mid Cu(s)$$

the standard emf of the cell is 2.70 V at 300 K. When the concentration of $Mg^{2\oplus}$ is changed to x M, the cell potential changes to 2.67 V at 300 K. The value of x is____.

(Given, $\frac{F}{R}$ = 11500 $KV^{-1}$, where F is the Faraday constant and R is the gas constant, ln(10) = 2.30)

**[JEE (Advanced) 2018]**

**CEC080**

**5.** For the following electrochemical cell at 298K,

$Pt(s) \mid H_2(g, 1bar) \mid H^{\oplus}(aq, 1M) \parallel M^{4\oplus}(aq), M^{2\oplus}(aq) \mid Pt(s)$

$E_{cell}$ = 0.092 V when $\frac{[M^{2\oplus}(aq.)]}{[M^{4\oplus}(aq.)]} = 10^x$

Given : $E^0_{M^{4\oplus}/M^{2\oplus}} = 0.151V$ ; $2.303\ \frac{RT}{F} = 0.059V$

The value of x is - **[JEE (Advanced) 2016]**

(A) –2 (B) –1 (C) 1 (D) 2

**CEC076**



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**6.** Some standard electrode potentials at 298 K are given below:

$Pb^{2\oplus}/Pb$ -0.13 V

$Ni^{2\oplus}/Ni$ -0.24 V

$Cd^{2\oplus}/Cd$ -0.40 V

$Fe^{2\oplus}/Fe$ -0.44 V

To a solution containing 0.001 M of $X^{2\oplus}$ and 0.1 M of $Y^{2\oplus}$, the metal rods X and Y are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of X. The correct combination(s) of X and Y, respectively, is (are)

(Given: Gas constant, R = 8.314 J $K^{-1}$ $mol^{-1}$, Faraday constant, F = 96500 C $mol^{-1}$)

**[JEE (Advanced) 2022]**

(A) Cd and Ni (B) Cd and Fe

(C) Ni and Pb (D) Ni and Fe

**CEC084**

**7.** Consider an electrochemical cell: A(s) | $A^{n\oplus}$ (aq, 2M) || $B^{2n\oplus}$ (aq, 1M) | B(s). The value of $\Delta H^0$ for the cell reaction is twice that of $\Delta G^0$ at 300 K. If the emf of the cell is zero, the $\Delta S^0$ (in $JK^{-1}$ $mol^{-1}$) of the cell reaction per mole of B formed at 300 K is___.

(Given : ln (2) = 0.7, R (universal gas constant) = 8.3 J $K^{-1}$ $mol^{-1}$. H, S and G are enthalpy, entropy and Gibbs energy, respectively.) **[JEE (Advanced) 2018]**

**CEC079**

**8.** The reduction potential ($E^0$, in V) of $MnO_4^{\ominus}$ (aq)/Mn(s) is _____. **[JEE (Advanced) 2022]**

[Given : $E^0_{(MnO_4^{\ominus}(aq)/MnO_2(s))} = 1.68V$ ; $E^0_{(MnO_2(s)/Mn^{2\oplus}(aq))} = 1.21V$ ; $E^0_{(Mn^{2\oplus}(aq)/Mn(s))} = -1.03V$ ]

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**CEC085**

**9.** The standard reduction potential data at 25°C is given below

E° ($Fe^{3\oplus}$, $Fe^{2\oplus}$) = +0.77 V ;

E° ($Fe^{2\oplus}$, Fe) = –0.44 V ;

E° ($Cu^{2\oplus}$, Cu) = +0.34 V ;

E° ($Cu^{\oplus}$, Cu) = +0.52 V ;

E° [$O_2(g) + 4H^{\oplus} + 4e^{\ominus} \rightarrow 2H_2O$] = +1.23 V ;

E° [$O_2(g) + 2H_2O + 4e^{\ominus} \rightarrow 4OH^{\ominus}$] = +0.40 V ;

E° ($Cr^{3\oplus}$, Cr) = –0.74 V ;

E° ($Cr^{2\oplus}$, Cr) = –0.91 V ;

Match E° of the redox pair in List-I with the values given in List-II and select the correct answer using the code given below the lists : **[JEE (Advanced) 2013]**

| List-I | List-II |
|---|---|
| (P) E°($Fe^{3\oplus}$, Fe) | (1) –0.18 V |
| (Q) E°($4H_2O \rightleftharpoons 4H^{\oplus} + 4OH^{\oplus}$) | (2) –0.4 V |
| (R) E°($Cu^{2\oplus} + Cu \rightarrow 2Cu^{\oplus}$) | (3) –0.04 V |
| (S) E°($Cr^{3\oplus}$, $Cr^{2\oplus}$) | (4) –0.83 V |

(A) P-4, Q-1, R-2, S-3 (B) P-2, Q-3, R-4, S-1

(C) P-1, Q-2, R-3, S-4 (D) P-3, Q-4, R-1, S-2

**CEC071**



For More Join: @IITJEE_Advanced

**10.** Plotting $1/\Lambda_m$ against $c\Lambda_m$ for aqueous solutions of a monobasic weak acid (HX) resulted in a straight line with y-axis intercept of P and slope of S. The ratio P / S is
[$\Lambda_m$ = molar conductivity
$\Lambda_m^\circ$ = limiting molar conductivity
c = molar concentration
$K_a$ = dissociation constant of HX] **[JEE (Advanced) 2023]**

(A) $K_a\ \Lambda_m^\circ$ (B) $K_a\ \Lambda_m^\circ\ /\ 2$

(C) $2\ K_a\ \Lambda_m^\circ$ (D) $1\ /\ (K_a\ \Lambda_m^\circ)$

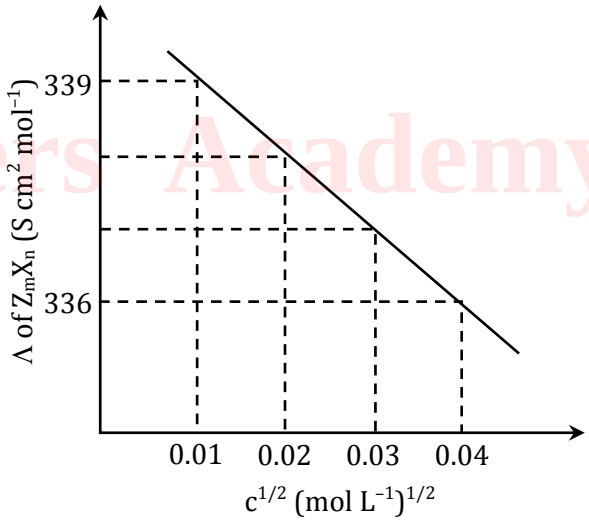
**CEC215**

**11.** Consider the strong electrolytes $Z_mX_n$, $U_mY_p$ and $V_mX_n$. Limiting molar conductivity ($\Lambda^0$) of $U_mY_p$ and $V_mX_n$ are 250 and 440 S cm$^2$ mol$^{-1}$, respectively. The value of (m + n + p) is _____.
Given :

| Ion | $Z^{n\oplus}$ | $U^{p\oplus}$ | $V^{n\oplus}$ | $X^{m\ominus}$ | $Y^{m\ominus}$ |
|---|---|---|---|---|---|
| $\lambda^0$(S cm$^2$ mol$^{-1}$) | 50.0 | 25.0 | 100.0 | 80.0 | 100.0 |

$\lambda^0$ is the limiting molar conductivity of ions

The plot of molar conductivity ($\Lambda$) of $Z_mX_n$ *vs* $c^{1/2}$ is given below. **[JEE (Advanced) 2022]**




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**CEC086**

**Paragraph for Q.12 and Q.13**

At 298 K, the limiting molar conductivity of a weak monobasic acid is $4 \times 10^2$ S cm$^2$ mol$^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation of $\alpha$ and the molar conductivity is $y \times 10^2$ Scm$^2$ mol$^{-1}$. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2$ S cm$^2$ mol$^{-1}$.

**12.** The value of y is _____. **[JEE (Advanced) 2021]**

**CEC082**

**13.** The value of $\alpha$ is _____. **[JEE (Advanced) 2021]**

**CEC083**



For More Join: @IITJEE_Advanced

**14.** The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 $cm^2$. The conductance of this solution was found to be $5 \times 10^{-7}$ S. The pH of the solution is 4. The value of limiting molar conductivity $\left(\Lambda_m^0\right)$ of this weak monobasic acid in aqueous solution is $Z \times 10^2$ S $cm^2$ $mol^{-1}$. The value of Z is.

**[JEE (Advanced) 2017]**

**CEC077**

**15.** The molar conductivity of a solution of a weak acid HX (0.01 M) is 10 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda_{X^\ominus}^0 \approx \lambda_{Y^\ominus}^0$, the difference in their $pK_a$ values, $pK_a(HX) - pK_a(HY)$, is (consider degree of ionization of both acids to be <<1).

**[JEE (Advanced) 2015]**

**CEC074**

**16.** In a conductometric titration, small volume of titrant of higher concentration is added stepwise to a larger volume of titrate of much lower concentration, and the conductance is measured after each addition.

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The limiting ionic conductivity $(\Lambda_0)$ values (in mS $m^2$ $mol^{-1}$) for different ions in aqueous solutions are given below :

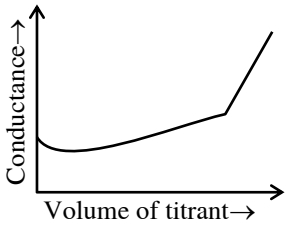
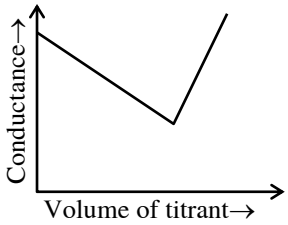
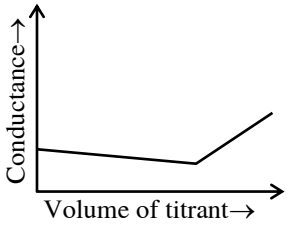
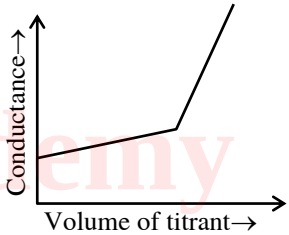
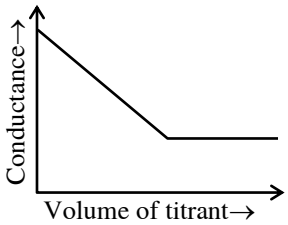
| Ions | $Ag^+$ | $K^+$ | $Na^+$ | $H^+$ | $NO_3^-$ | $Cl^-$ | $SO_4^{2-}$ | $OH^-$ | $CH_3COO^-$ |
|---|---|---|---|---|---|---|---|---|---|
| $\Lambda_0$ | 6.2 | 7.4 | 5.0 | 35.0 | 7.2 | 7.6 | 16.0 | 19.9 | 4.1 |

For different combinations of titrates and titrants given in **List-I**, the graphs of 'conductance' versus 'volume of titrant' are given in **List-II**.

Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.



For More Join: @IITJEE_Advanced

| List-I | List-II |
|---|---|
| (P) Titrate : KCl<br>Titrant : $AgNO_3$ | (1)  |
| (Q) Titrate : $AgNO_3$<br>Titrant : KCl | (2)  |
| (R) Titrate : NaOH<br>Titrant : HCl | (3)  |
| (S) Titrate : NaOH<br>Titrant : $CH_3COOH$ | (4)  |
| | (5)  |



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(A) P → 4, Q → 3, R → 2, S → 5
(B) P → 2, Q → 4, R → 3, S → 1
(C) P → 3, Q → 4, R → 2, S → 5
(D) P → 4, Q → 3, R → 2, S → 1

[JEE (Advanced) 2024]
CEC287



For More Join: @IITJEE_Advanced

**17.** An aqueous solution of X is added slowly to an aqueous solution of Y as shown in List-I. The variation in conductivity of these reactions is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists : **[JEE (Advanced) 2013]**

| List-I | List-II |
|---|---|
| (P) $(C_2H_5)_3N + CH_3COOH$ (X, Y) | (1) Conductivity decreases and then increases |
| (Q) $KI(0.1M) + AgNO_3(0.01M)$ (X, Y) | (2) Conductivity decreases and then does not change much |
| (R) $CH_3COOH + KOH$ (X, Y) | (3) Conductivity increases and then does not change much |
| (S) NaOH + HI (X, Y) | (4) Conductivity does not change much and then increases |

(A) P-3, Q-4, R-2, S-1 (B) P-4, Q-3, R-2, S-1 (C) P-2, Q-3, R-4, S-1 (D) P-1, Q-4, R-3, S-2

**CEC072**

**18.** An aqueous solution of hydrazine ($N_2H_4$) is electrochemically oxidized by $O_2$, thereby releasing chemical energy in the form of electrical energy. One of the products generated from the electrochemical reaction is $N_2(g)$.

Choose the correct statement(s) about the above process :

(A) $OH^-$ ions react with $N_2H_4$ at the anode to form $N_2(g)$ and water, releasing 4 electrons to the anode.

(B) At the cathode, $N_2H_4$ breaks to $N_2(g)$ and nascent hydrogen released at the electrode reacts with oxygen to form water.

(C) At the cathode, molecular oxygen gets converted to $OH^-$.

(D) Oxides of nitrogen are major by-products of the electrochemical process.

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**[JEE (Advanced) 2024]**

**CEC288**

**19.** All the energy released from the reaction $X \rightarrow Y$, $\Delta_r G^\circ = -193$ kJ $mol^{-1}$ is used for the oxidizing $M^{\oplus}$ and $M^{\oplus} \rightarrow M^{3\oplus} + 2e^{\ominus}$, $E^\circ = -0.25$ V.

Under standard conditions, the number of moles of $M^{\oplus}$ oxidized when one mole of X is converted to Y is : [F = 96500 C $mol^{-1}$] **[JEE (Advanced) 2015]**

**CEC075**



For More Join: @IITJEE_Advanced

# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | D | B | A | B | A | A | B | D | A | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | D | C | B | C | C | C | A | B | A |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | A | A | D | C | D | C | A | B | A | C |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | |
| Ans. | C | A | D | B | B | B | D | D | B | |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | AD | C | D | A | D | A | C | C | A |

## EXERCISE (S)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** | 2 | **2.** | 3 | **3.** | 26 | **4.** | 3 | **5.** | 2500 |
| **6.** | 125 | **7.** | 9 | **8.** | 10 | **9.** | 150 | **10.** | 400 |
| **11.** | 60 | **12.** | 4 | **13.** | 2 | **14.** | 0.78 | | |
| **15.** | 0.53,0.38 | **16.** | 0.80 | **17.** | 0.05 or 0.06 | **18.** | 0.41 | **19.** | 3.86 |
| **20.** | 7.60 | **21.** | 0.59 | **22.** | 250.00, 125.00 | **23.** | –0.16 | **24.** | 9.40 |
| **25.** | (a) 0.10 (b) 0.03 (c) 0.30 | | | **26.** | (a) 0.75 ; (b) 0.69 ; (c) 1.10 | | | | |

Bankers Academy JEE

## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 2 | 1 | 2 | 3 | 3 | 2 | 4 | 18 | 3 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 275 | 3 | 917 | 1 | 2 | 1 | 3 | 338 OR 339 | 17 | 6 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | 1 | 150 | 2 | 4 | 25 | 3 | 4 | 1 | 2 | 3 |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | 1 | 25 | 66 | 2 | 2 | 3 | 4 | 4 | 3 | 2 |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | 107 gm or 108 | 45.75 Or 46 | 11 | 5 | 161 | 3 | 4 | 1 | 2 | 1 |
| Que. | 51 | 52 | | | | | | | | |
| Ans. | 4 | 3 | | | | | | | | |

## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | AB | 13.32 | B | 10 | D | ABC | –11.62 | 0.77 | D | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | |
| Ans. | 7 | 0.86 | 0.21 or 0.22 | 6 | 3 | C | A | AC | 4 | |



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